

Recursive Estimation	
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1 Probability Theory	
1.1 Random Variables	
1.1.1 Discrete Random Variables	
DRV – Discrete Random Variable defined by $\mathcal{X}, p_x: \mathcal{X} \rightarrow \mathbb{R}$	
$\mathcal{X} \subset \mathbb{Z}$ of all possible outcomes $p_x(\bar{x}) \geq 0$ and $\sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x}) = 1$ or $\sum_{\bar{x} \in \mathcal{X}} p_{x y}(\bar{x} \bar{y}) = 1$	
Margin. / Sum Rule	$p(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}} p_{xy}(\bar{x}, \bar{y})$
Cond. / Product Rule	$p_{x y}(\bar{x} \bar{y}) := \frac{p_{xy}(\bar{x}, \bar{y})}{p_y(\bar{y})}$
Total Prob. Theorem	$p_x(\bar{x}) := \sum_{\bar{y} \in \mathcal{Y}} p_{x y}(\bar{x} \bar{y})p_y(\bar{y})$
1.1.2 Continuous Random Variables	
CRV – Continuous Random Variable: DRV $\sum \rightsquigarrow$ CRV $\int$	
- Set $\mathcal{X} \subset \mathbb{R}$ section of real line - PDF p_x satisfies $p_x(\bar{x}) \geq 0$ and $\int_{\mathcal{X}} p_x(\bar{x})d\bar{x} = 1$	
Caution: $Pr(x = \bar{x}) = 0, \quad Pr(x \in (a, b)) = \int_a^b p(\bar{x})d\bar{x}$	
1.1.3 Conditional PDF	
Conditional PDF	
Margin. / Sum Rule	$p(\bar{x} \bar{z}) = \sum_{\bar{y} \in \mathcal{Y}} p_{xy z}(\bar{x}, \bar{y} \bar{z})$
Cond. / Product Rule	$p_{x yz}(\bar{x} \bar{y}, \bar{z}) := \frac{p_{xy z}(\bar{x}, \bar{y} \bar{z})}{p_{y z}(\bar{y} \bar{z})}$
Independence	
$p(x y) = p(x) \Leftrightarrow p(y x) = p(y) \Leftrightarrow p(x, y) = p(x)p(y)$	
$p(x, y, z) = p(x, y z)p(z) \rightarrow x, y$ indep. $p(x, y z) = p(z x, y)$	
Conditional Independence	
The knowledge of z makes x and y independent:	
$p(x y, z) = p(x z) \Leftrightarrow p(x, y z) = p(x z)p(y z)$	
Caution!!!! in general we still have: $p(x, y) \neq p(x)p(y)$	
Caution!!!! Independence $\nRightarrow$ Conditional Independence	
1.2 Expectation and Variance	
1.2.1 Expectation	
Definition: Integral for CRV	
$\mathbb{E}_x[x] = \sum_{\bar{x} \in \mathcal{X}} \bar{x}p_x(\bar{x})$	
- Linearity: $\mathbb{E}_{xy}[a + bx + cy] = a + b\mathbb{E}_x[x] + c\mathbb{E}_y[y]$ - Multi Variable: $\mathbb{E}_{xy}[g(x, y)] = \sum_{\bar{y}} \sum_{\bar{x}} g(\bar{x}, \bar{y})p_{xy}(\bar{x}, \bar{y})$ - Independence of $x, y \Rightarrow \mathbb{E}_{xy}[xy] = \mathbb{E}_x[x]\mathbb{E}_y[y]$	
Law of Unconscious Statistician for $y = g(x)$	
$\mathbb{E}_y[y] = \sum_{\bar{y} \in \mathcal{Y}} \bar{y}p_y(\bar{y}) = \sum_{\bar{x} \in \mathcal{X}} g(\bar{x})p_x(\bar{x})$	
1.2.2 Variance (generally a matrix)	
$\text{Var}_x[x] = \mathbb{E}_x[(x - \mathbb{E}_x[x])(x - \mathbb{E}_x[x])^\top] \stackrel{1D}{=} \mathbb{E}_x[x^2] - \mathbb{E}_x[x]^2$	
Linearity	
$\text{Var}_x[a + bX + cY] = b^2 \text{Var}_X[x] + c^2 \text{Var}_Y[y] + 2bc \text{Cov}[X, Y]$	
Covariance: if independent $\text{Cov}(X, Y) = 0$	
$\text{Cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$	
1.3 Distribution Sampling	
CDF – Cumulative Distribution Function	
$\hat{F}_x(\bar{x}) := \sum_{i=-\infty}^{\bar{x}} \hat{p}_x(i) = Pr(x \leq \bar{x}) = \int_{-\infty}^{\bar{x}} p_x(\hat{x})d\hat{x}$	
A1 – 1 DRV $\bar{u}$ sample of $u \sim \mathcal{U}$, solve for $\bar{x}$ such that:	
$\hat{F}_x(\bar{x} - 1) < \bar{u} \leq \hat{F}_x(\bar{x})$	
A2 – Multiple DRV or CRV	
Works for finite & infinite # of elements	

- Decompose $\hat{p}(\bar{x}, \bar{y}) = \hat{p}_{x y}(\bar{x} \bar{y})\hat{p}_y(\bar{y})$ - Apply A1 to get sample $\bar{y}$ via $\hat{p}_y(\bar{y})$ - With $\bar{y}$ fixed, apply A1 to get sample $\bar{x}$ via $\hat{p}_{x y}(\bar{x} \bar{y})$	
A3 – 1 CRV $\hat{p}_x$: pw-conti., bounded. $u \sim \mathcal{U}(0, 1)$ and solve:	
$\bar{u} = F_x(\bar{x}) \rightarrow \bar{x} = F_x^{-1}(\bar{u})$	
1.4 Change of Variables	
DRV: $x = g(y)$	
$p_x(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}: g(\bar{y}) = \bar{x}} p_y(\bar{y})$	
CRV general case $y = g(x), \rightarrow x = g^{-1}(y)$:	
$p_y(\bar{y}) = \frac{p_x(\bar{x})}{\left \frac{dg}{dx}(\bar{x})\right }, \quad \bar{x} = g^{-1}(y)$	
CRV – Multivar. & Indep. Case: $z = x + y$	
$p_z(\bar{z}) = \sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x})p_y(\bar{z} - \bar{x})$	
CRV – Multivar. & Cond. Case: $z = g(w, x), w = h(z, x)$	
Assumptions: - $g(w, x), p_w$ cont. diff'bar	
- $g(w, x)$ strictly monotonic	
- For scalars $\det(x)$ is just x	
$p_{z x}(\bar{z} \bar{x}) = p_w(h(\bar{z}, \bar{x}) \bar{x}) \cdot \det(\frac{\partial g}{\partial w}g(h(\bar{z}, \bar{x}), \bar{x})) ^{-1}$	
1.5 Miscellaneous	
Indep. multiplication: $z = xy \rightarrow p(z) = \sum_Y p_x(\frac{z}{y})p_y(y)$	
Uniform Distribution	
$p(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{else} \end{cases} \quad \left \quad \begin{aligned} \mathbb{E}\{x\} &= \frac{a+b}{2} \\ \text{Var}\{x\} &= \frac{(b-a)^2}{12} \end{aligned} \right.$	
Triangular Distribution	
$p(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & a \leq x \leq c \\ \frac{2(b-x)}{(b-a)(b-c)} & c \leq x \leq b \\ 0 & \text{else} \end{cases}$	
$F(x) = \begin{cases} 0 & x < a \\ \frac{(x-a)^2}{(b-a)(c-a)} & a \leq x \leq c \\ 1 - \frac{(b-x)^2}{(b-a)(b-c)} & c \leq x \leq b \\ 1 & x > b \end{cases}$	
$\mathbb{E}\{x\} = \frac{a+b+c}{3}, \quad \text{Var}\{x\} = \frac{a^2+b^2+c^2-ab-ac-bc}{18}$	
Gaussian Distribution	
PDF Gaussian distributed vector CRV $y = (y_1, \dots, y_D)$	
$p(x) = \frac{1}{(2\pi)^{D/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{(x - \mu)^T \Sigma^{-1} (x - \mu)}{2}\right)$	
$\rightsquigarrow$ fully characterized by mean $\mu \in \mathbb{R}^D$ and var $\Sigma \in \mathbb{R}^{D \times D}$	
(Scalar: $D = 1, \sigma^2 = \det(\Sigma) \rightarrow \det(\Sigma)^{1/2} = \sigma$)	
$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad \left \quad \begin{aligned} \mathbb{E}\{x\} &= \mu \\ \text{Var}\{x\} &= \sigma^2 \end{aligned} \right.$	
Sum of two independent Gaussians:	
$x_1 \sim \mathcal{N}(\mu_1, \Sigma_1), x_2 \sim \mathcal{N}(\mu_2, \Sigma_2) \rightarrow z = M_1x_1 + M_2x_2$	
$z \sim \mathcal{N}(M_1\mu_1 + M_2\mu_2, M_1\Sigma_1M_1^T + M_2\Sigma_2M_2^T)$	
Diagonal Variance with $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_D^2), \mu = [\mu_D^1]$	
$\rightsquigarrow y_i$ mut.indep. iff Σ diag.	
$p(y) = \prod_{i=1}^D \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(y_i - \mu_i)^2}{2\sigma_i^2}\right)$	
Dependent Gaussians:	
$\Sigma_{x \pm y} = \Sigma_x + \Sigma_y \pm \Sigma_{xy} \pm \Sigma_{xy}^T, \mu_{x \pm y} = \mu_x \pm \mu_y$	

2 Bayesian Tracking	
2.1 Bayes Theorem	
Definition: (Applies to discrete and continuous random variables)	
$p(x y) = \frac{p(y x)p(x)}{p(y)}$	
x =system state, $p(x)$ =prior belief of state, z =observations related to state $p(z x)$ =observation likelihood, $p(x z)$ =posterior belief of state	
$p(z)$ =normalization constant $= \sum_x p(z x)p(x)$	
For multiple observations:	
$p(x z_1, \dots, z_N) = \underbrace{\sum_{x \in \mathcal{X}} p(x)}_{\text{normalization}} \underbrace{\overbrace{p(x)}^{\text{prior}} \prod_i \overbrace{p(z_i x)}^{\text{observation likelihood}}}_{\text{likelihood}}$	
2.2 Bayesian Tracking	
Model	
Objective: Calculate pdf of state $p(x(k) z(1:k))$	
$x(k) = q_{k-1}(x(k-1), v(k-1)) \in \mathcal{X}$	
$z(k) = h_k(x(k), w(k))$	
Prior Update - State Prediction	
$p(x(k) z(1:k-1)) = \sum_{x(k-1)} \underbrace{p(x(k) x(k-1))}_{\text{process model}} \underbrace{p(x(k-1) z(1:k-1))}_{\text{previous iteration}}$	
Measurement Update	
$p(\bar{x}(k) \bar{z}(1:k)) = \frac{\overbrace{p(\bar{z}(k) \bar{x}(k))}^{\text{measurement model}} \overbrace{p(\bar{x}(k) \bar{z}(1:k-1))}^{\text{prior}}}{\underbrace{\sum_{i \in \mathcal{X}} p(\bar{z}(k) i)p(i \bar{z}(1:k-1))}_{\text{normalization}})}$	
Computer Implementation	
1. Enumerate state space $\mathcal{X} = \{0, 1, \dots, N-1\}$	
2. Posterior define $\mathbf{a}_{k k}^i := p_{x(k) z(1:k)}(i \bar{z}(1:k))$	
3. Prior define $\mathbf{a}_{k k-1}^i := p_{x(k) z(1:k-1)}(i \bar{z}(1:k-1))$	
Initialization $\mathbf{a}_{0 0} = p_{x(0)}(i) \quad i = 0, \dots, N-1$	
$\left. \begin{aligned} \mathbf{a}_{k k-1}^i &= \sum_{j=0}^{N-1} p_{x(k) x(k-1)}(i j) \mathbf{a}_{k-1 k-1}^j \\ \mathbf{a}_{k k}^i &= \frac{p_{z(k) x(k)}(\bar{z}(k) i) \mathbf{a}_{k k-1}^i}{\sum_{j=0}^{N-1} p_{z(k) x(k)}(\bar{z}(k) j) \mathbf{a}_{k k-1}^j} \end{aligned} \right\} \text{Iterate } k > 0$	
3 Extracting Estimates	
3.1 General Measurement	
$\underbrace{\begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}}_{z \in \mathbb{R}^m} = \underbrace{\begin{bmatrix} h_{11} & h_{1n} \\ \vdots & \vdots \\ h_{m1} & h_{mn} \end{bmatrix}}_{H \in \mathbb{R}^{m \times n}} \cdot \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x \in \mathbb{R}^n} + \underbrace{\begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}}_{w \in \mathbb{R}^m}$	
m Measurements, n States, $w_i \sim \mathcal{N}(0, R)$ – Zero-mean additive gaussian, $x, w_i \forall i$ assumed mutually independent	
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3.3 Likelihood Estimators	
3.3.1 ML-Maximum Likelihood	
Choose x to make $\bar{z}$ most likely:	
$\underbrace{x \in \mathcal{X}}_{\text{unknown}} \left\{ \quad \underbrace{p_x(\bar{x})}_{\text{unknown}} \quad \right\} \quad \hat{x}^{ML} := \underset{\bar{x} \in \mathcal{X}}{\text{argmax}} [p_{z x}(\bar{z} \bar{x})]$	
$p_{z x}(\bar{z} \bar{x})$ is var. – technically not a PDF	
3.3.2 MAP-Maximum A Posteriori	

ML given observation & prior belief about x :	
$\underbrace{x \in \mathcal{X}}_{p_x(\bar{x})} \left\{ \quad \underbrace{\text{unknown}}_{\text{known}} \quad \right\} \quad \hat{x}^{MAP} := \underset{\bar{x} \in \mathcal{X}}{\text{argmax}} [p_{z x}(\bar{z} \bar{x})p_x(\bar{x})]$	
$\hat{x}^{MAP} = \hat{x}^{ML}$ for constant $p_x(\bar{x})$	
3.4 General Measurement	
$\underbrace{\begin{bmatrix} z_1 \\ \vdots \\ z_m \end{bmatrix}}_{z \in \mathbb{R}^m} = \underbrace{\begin{bmatrix} h_{11} & h_{1n} \\ \vdots & \vdots \\ h_{m1} & h_{mn} \end{bmatrix}}_{H \in \mathbb{R}^{m \times n}} \cdot \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x \in \mathbb{R}^n} + \underbrace{\begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix}}_{w \in \mathbb{R}^m}$	
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ML given observation & prior belief about x :	
$\underbrace{x \in \mathcal{X}}_{p_x(\bar{x})} \left\{ \quad \underbrace{\text{unknown}}_{\text{known}} \quad \right\} \quad \hat{x}^{MAP} := \underset{\bar{x} \in \mathcal{X}}{\text{argmax}} [p_{z x}(\bar{z} \bar{x})p_x(\bar{x})]$	
$\hat{x}^{MAP} = \hat{x}^{ML}$ for constant $p_x(\bar{x})$	
3.6 Least Squares Estimators	
3.6.1 Linear Least Squares	
ML estimate when errors are indep. 0-mean, smae variance, $\mathcal{N}$ -distributed:	
$\hat{x}^{LS} := \underset{x}{\text{argmin}} \left[(z - Hx)^\top (z - Hx) \right] = (H^\top H)^{-1} H^\top z$	
$\hat{x}^{LS} = \hat{x}^{ML}$ for gaussian noise	

3.6.2	Weighted Least Squares $\hat{x}^{WLS} := \operatorname{argmin}_x \left[(\bar{z} - H\hat{x})^\top R^{-1} (\bar{z} - H\hat{x}) \right]$ $= (H^\top R^{-1} H)^{-1} H^\top R^{-1} \bar{z}$ Incorporate Prior Knowledge on x: $r \sim \mathcal{N}(0, P_x) \Rightarrow x := \hat{x} + r, \mathbb{E}\{x\} = \hat{x}_0, \operatorname{Var}\{x\} = P_x$ Introduce Extended System: $\tilde{z} = \begin{bmatrix} \hat{x}_0 \\ z(1:k) \end{bmatrix}, \tilde{H} = \begin{bmatrix} \mathbb{I} \\ H(1:k) \end{bmatrix} \tilde{w} = \begin{bmatrix} -r \\ w(1:k) \end{bmatrix}, \tilde{R} = \begin{bmatrix} P_x & 0 \\ 0 & R(1:k) \end{bmatrix}$ 3.6.3 Recursive Least Squares Interpretation Update $\hat{x}$ recursively when new $\bar{z}(k)$ arrives Estimation Error: $e(k) = (\mathbb{I} - KH) e(k-1) - Kw(k)$ Objective Minimize MSE with $P(k) = \operatorname{Var}\{e(k)\}$ $J(k) := \mathbb{E} \left\{ e^\top(k) e(k) \right\} = \mathbb{E} \{ \operatorname{trace}(P(k)) \}$ Algorithm Initialization: $\hat{x}(0) = \hat{x}_0, \quad P(0) = P_x = \operatorname{Var}\{x\}$ Recursion: Observe $\bar{z}(k)$, then update: $K(k) = P(k-1)H(k)^\top \left(H(k)P(k-1)H(k)^\top + R(k) \right)^{-1}$ $\hat{x}(k) = \hat{x}(k-1) + K(k) (\bar{z}(k) - H(k)\hat{x}(k-1))$ $P(k) = (\mathbb{I} - K(k)H(k))P(k-1)(\mathbb{I} - K(k)H(k))^\top + K(k)R(k)K(k)^\top$ <table><tr><td>4</td><td>Kalman Filter</td></tr><tr><td>5</td><td>Particle Filter</td></tr><tr><td>6</td><td>Observer Based Control</td></tr></table>	4	Kalman Filter	5	Particle Filter	6	Observer Based Control
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